Experiment = To determine the modulus of rigidity of a wire by Maxewell's needle (Dynamic method)

Apparatus =

- Maxewell's needle
- a wire of suitable length and diameter
- stop watch
- torsion head
- telescope
- knitting needle fitted vertically on a stand
- screw guage
- spring balance
- meter rod

Diagram =

Observation and Calculation =

length of given wire = L = ... 100.33 ... cm

Total length of maxwell's needle = 2l = ... 12 ... cm

Half length of needle = l = ... 6 cm

Average mass of hollow cylinder = m_1 = ... 12.5 ... cm

Average mass of solid cylinder = m_2 = ... 162.5 ... cm

L.C of screw guage = ... 0.01 ... mm ... 0.001 ... cm

Zero Correction of screw guage = ... Null ... mm ... Null ... cm

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APPARATUS :-

- Maxewell's needle • a wire of suitable length and diameter • Stopwatch • torsion head • meter rod
- knitting needle fitted vertically on a stand
- Telescope • meter rod • Spring balanced

Procedure :-

- Take a wire of suitable length and diameter. Remove the kniks by stretching the wire gently. Suspend the maxewells needle from one end of wire and fixed other end with rigid support.
- Arrange the apparatus so that mirror attaches to middle of maxewells needle faces towards a window and this needles hangs horizontally parallel to wall carrying support.
- Place the knitting needle on an upright in front of mirror.
- Place telescope in front of mirror atleast 5 meter away from it and focus it at image of knitting needle.
- Put the holow cylinder of maxewell's needle inwards and solid cylinder outwards systematically in the tube.
- Displace one end of Maxewell's table slightly backwords to make it vibrate in a horizontal Plane.
- Start the stopwatch when image of needle just passes through vertical - cross wire. Count 1 when image

Corrected diameter of wire = 1) 0.041 cm   2) 0.042 cm   3) 0.0412 cm

Mean diameter of wire = d = 0.041 cm

Radius of wire = $r = \frac{d}{2}$ = 0.0205 cm

Position of solid cylinder	No of Oscillation	Time for 20 Vibrations 1 sec	2 sec	3 sec	Mean t sec	Time period = T, T = t/20 sec
Outside	20	191	192	191.5	191.5	9.56
Inside	20	161	160	161	160.2	8.01

Calculation = Modulus of rigidity = $\frac{8\pi l (m_2 - m_1) L^2}{(T_1^2 - T_2^2) r^4}$

$$= \frac{8(3.14)(100.3)(162.5 - 12.5)(6)^2}{[(9.56)^2 - (8.01)^2](0.0205)^4} = 7.1 \times 10^{11} \text{ dynes/cm}^2$$

Modulus of rigidity of given wire = 7.1×10^{11} dynes/cm²

Actual Value of η = 8.9×10^{11} dynes/cm²

Percentage error = $\frac{\text{Actual Value} - \text{Calculated Value}}{\text{Actual Value}} \times 100$

= 20 %

crosses the vertical cross wire again.

- Now repeat the experiment by interchanging both cylinders inwards and both hollow cylinder outwards.
- Again find time for 20 vibrations and repeat it thrice.
- Measure the length of given wire b/w to torsion heads and length of maxewell's wire.
- Measure the masses of both hollow cylinder's and find out average mass m_1 of each hollow cylinder's. Same process for measuring mass m_2 of each solid cylinder
- Calculate the modulus of rigitidy:

$$\eta = \frac{8\pi l\,(m_2 - m_1)L^2}{(T_1^2 - T_2^2)\; r^4}$$

Precautions :-

- The wire free from kinks should be used.
- The amplitude of maxewell's needle should be small.
- The maxewell's needle should vibrate equallyon both sides of reference needle and no up and down movement.
- Measure diameter of wire at various points
- The needle should be protected from air-currents during performance of experiment.